

Momentum and Centre of Mass

Building Models to Describe our World
Systems, Collisions and Centre of Mass



Outline

Momentum Basics

- Momentum of a particle

- Impulse

- Systems of particles

- Momentum of a system

Collisions

- Concept

- Elastic vs inelastic

- Elastic collision in 1D

- How to identify an elastic collision

- Example: basketball and tennis ball

- Ballistic pendulum

Centre of Mass

- Describing a system

- Concept

 - CM location

 - CM velocity

 - CM reference frame

- CM for a continuous object

- CM of a square

- CM of a triangle

- CM of a 3D object

Momentum Basics

Momentum and Centre of Mass

Momentum of a particle

- The momentum of a particle of mass m is:

$$\vec{p} = m\vec{v}$$

- The rate of change of momentum is the net force on the particle:

$$\begin{aligned}\frac{d\vec{p}}{dt} &= \frac{d}{dt}m\vec{v} = m\frac{d\vec{v}}{dt} = m\vec{a} \\ \therefore \frac{d\vec{p}}{dt} &= \sum \vec{F} = \vec{F}^{net}\end{aligned}$$

- **If the net force on a particle is zero, its momentum is constant** (it doesn't change with time).

Impulse

- The net impulse **vector** is the change in momentum:

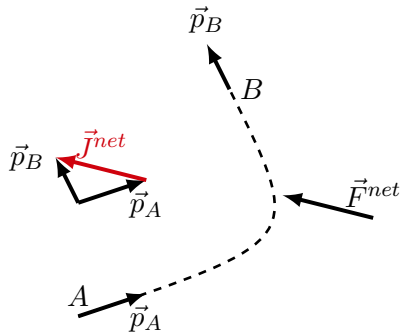
- If force does not change with time:

$$\frac{d\vec{p}}{dt} \sim \frac{\Delta\vec{p}}{\Delta t} = \vec{F}^{net}$$
$$\therefore \vec{J}^{net} = \Delta\vec{p} = \vec{F}^{net} \Delta t$$

- If force changes magnitude or direction with time

$$\vec{J}^{net} = \Delta\vec{p} = \int_{t_a}^{t_b} \vec{F}^{net}(t) dt$$

- A force acting for a given time will change the momentum.
- What makes it different from Work?*



Impulse is the change in momentum.

$$\vec{J}^{net} = \Delta\vec{p} = \vec{p}_B - \vec{p}_A = \vec{F}^{net} \Delta t$$

Systems of particles

- Newton's Second Law is only valid for a point particle (an infinitely small particle of mass, m).
- Loosely, we call a collection of particles a “system”.
- A solid block of matter is a system of many particles that cannot move relative to each other.
- A gas is a system of many particles that can move relative to each other.
- So far, we have modelled, say, blocks and treated them like point particles even if they are not.
- We can distinguish between “internal” and “external” forces to the system.

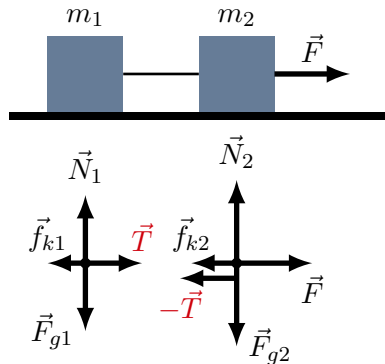
Modeling 2 blocks as a system

- Want to know the acceleration of the two blocks, given F , friction.
- Can apply Newton's Second Law to each block (4 equations, blocks have the same acceleration):

$$\sum F_1^{horiz} = T - \mu_k N_1 = m_1 a$$

$$\sum F_2^{horiz} = F - T - \mu_k N_2 = m_2 a$$

+ forces in vertical direction



Modeling two connected blocks, either individually or as a system.

Modeling 2 blocks as a system

- But, the motion of the system, $(m_1 + m_2)$, depends on the external forces on the system. Tension is an *internal* force.
- Consider adding together Newton's Second Law for each particle:

$$\sum F_1^{\text{horiz}} = T - \mu_k N_1 = m_1 a$$

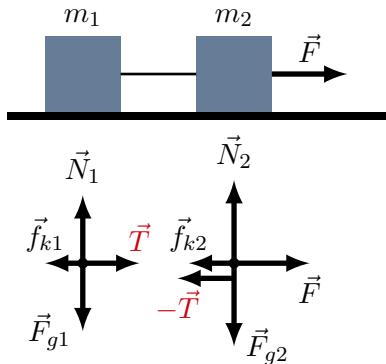
$$\sum F_2^{\text{horiz}} = F - T - \mu_k N_2 = m_2 a$$

$$\sum F_1^{\text{horiz}} + \sum F_2^{\text{horiz}} = \textcolor{red}{T} - \mu_k N_1 + F - \textcolor{red}{T} - \mu_k N_2$$

Modeling two connected blocks, either individually or as a system.

$$\therefore F - \mu_k (N_1 + N_2) = (m_1 + m_2) a$$

$$F - \mu_k (m_1 + m_2) g = (m_1 + m_2) a$$



The internal forces (exerted by one object of the system on another) always cancel out because of Newton's Third Law. Only the external forces affect the motion of the system.

Momentum of a system

- If many particles make up a system, we can add their individual momenta together to define the total momentum of the system:

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots$$

- The rate of change of momentum of the system is given by the sum of the **external forces** exerted on the system:

$$\frac{d\vec{P}}{dt} = \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} + \frac{d\vec{p}_3}{dt} + \dots = \sum \vec{F}^{ext}$$

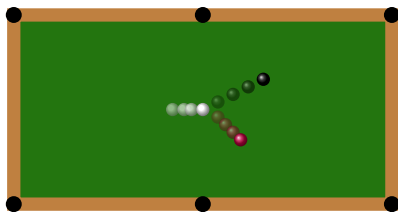
- If there is **no net external force on the system**, the momentum of the system **is conserved**.

Collisions

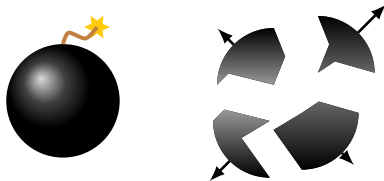
Momentum and Centre of Mass

Collisions

- “Events” where one considers the total momentum of a system “just before” and “just after”.
- Examples:
 - Billiard balls colliding.
 - An arrow embedding itself into a block of wood that can move.
 - A bomb exploding into fragments.
- Even if there are net external forces (e.g. gravity), one can **usually** neglect them if the momentum is compared just before and just after the collision and model total momentum as being conserved.



Billiard balls colliding.



An explosion.

Elastic vs Inelastic

- Just because momentum is conserved does not mean that mechanical energy is conserved:
 - **Momentum** is conserved when there is **no net external force** on the system.
 - **Energy** is conserved when there is **no non-conservative force** doing work on any particle in the system.
- To model elastic collisions, the mechanical energy of the system is conserved before and after the collision:

$$E = K + U = K' + U' = E'$$

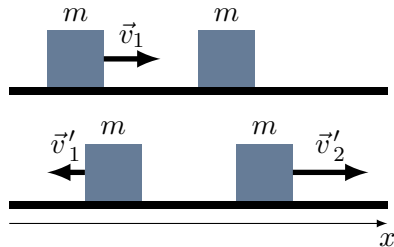
- K, U are the total kinetic energy and total potential energy of the system (add together the kinetic and potential energies of all particles in the system).
- Usually, if the collision happens over a small distance, then the potential energies are (almost) the same before and after the collision (potential energy depends on position and if position doesn't change much, neither does U):

$$U = U' \quad (\text{Usually})$$

Elastic collision in 1D: m collides with m at rest

- Conservation of momentum:

$$m\vec{v}_1 = m\vec{v}'_1 + m\vec{v}'_2$$



A block of mass, m , with speed v_1 , collides elastically with another block of mass, m , initially at rest.

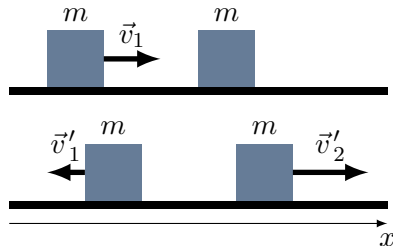
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- x -component:

$$v_1 = v'_1 + v'_2$$



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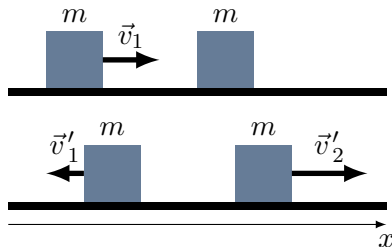
$$v_1 = v'_1 + v'_2$$

- Conservation of energy:

$$\frac{1}{2}mv_1^2 = \frac{1}{2}mv'^2_1 + \frac{1}{2}mv'^2_2$$

$$\therefore v_1^2 = v'^2_1 + v'^2_2$$

- 2 equations 2 unknowns, but not trivial, since not linear (i.e. quadratic)



A block of mass, m , with speed v_1 , collides elastically with another block of mass, m , initially at rest.

Useful trick for elastic collisions

- Instead of writing “before”= “after”:

$$v_1 = v_1' + v_2'$$
$$v_1^2 = v_1'^2 + v_2'^2$$

- Write “1” = “2”:

$$v_1 - v_1' = v_2' \quad (1)$$

$$v_1^2 - v_1'^2 = v_2'^2 \quad (2)$$

$$\therefore (v_1 - v_1')(v_1 + v_1') = v_2'^2 \quad (\text{factoring equation 2}) \quad (3)$$

- Divide equation 3 by equation 1:

$$\frac{(v_1 - v_1')(v_1 + v_1')}{v_1 - v_1'} = v_2' \quad \rightarrow \quad \boxed{v_1 + v_1' = v_2'}$$

Elastic collision in 1D: m collides with m at rest

- Solve for the final velocities:

$$v_1 - v_1' = v_2'$$

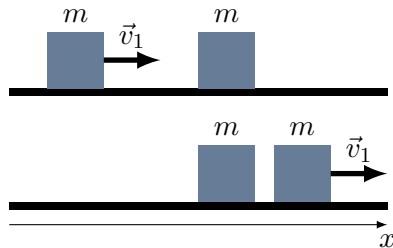
$$v_1 + v_1' = v_2'$$

- Which gives:

$$v_1' = 0$$

$$v_2' = v_1$$

The first block comes to rest and the second one moves with the same speed as the first one did!



A block of mass, m , with speed v_1 , collides elastically with another block of mass, m , initially at rest.

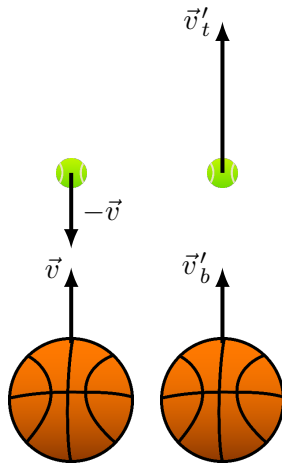
How to tell if it is elastic?

- Calculate the initial and final kinetic energies and compare them.
 - Problem solving hint: If you are given enough information to be able to calculate the initial and final velocities using only conservation of momentum, chances are it was inelastic...
- If two objects stick together, or come unstuck (e.g. explosion), then the energy is not conserved:
 - Think of two equal masses coming towards each other at the same speed and coming to rest stuck together. The total momentum is always zero, but the total kinetic energy went from non-zero to zero.
- If you are told about the collision mechanism, you might be able to determine if it's elastic. E.g. colliding with a spring is elastic (that's where the term comes from).

Basketball and tennis ball

- Just before the collision, assume that tennis ball speed = basketball speed. Assume elastic collision (between balls and with the ground).
- Work it out... you will find:

$$v'_t = \frac{3M - m}{M + m}v$$



A basketball colliding with a tennis ball.

Basketball and tennis ball

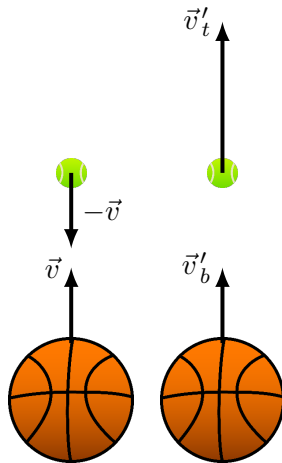
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- Re-write as (divide through by M):

$$v'_t = \frac{3 - \frac{m}{M}}{1 + \frac{m}{M}}v$$

- If $M \gg m$, $\frac{m}{M} \sim 0$ and $v'_t \sim 3v$.

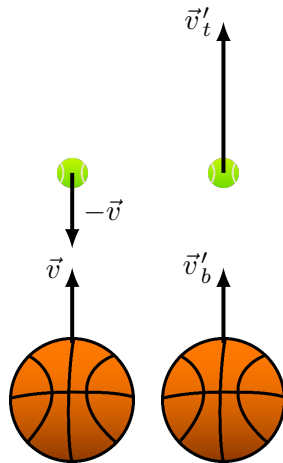


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Basketball and tennis ball

- If $M \gg m$, $\frac{m}{M} \sim 0$:

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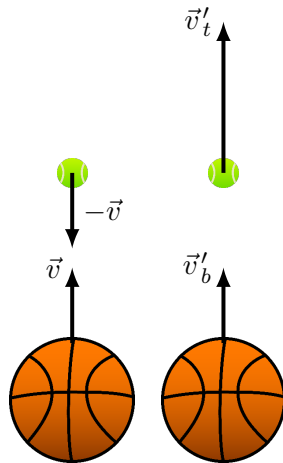
- If $M \gg m$, $\frac{m}{M} \sim 0$:

$$v'_t = \frac{3 - \frac{m}{M}}{1 + \frac{m}{M}} v \sim 3v$$

- With speed v , the tennis ball will go a height h :

$$mgh = \frac{1}{2}mv^2$$
$$\therefore h \propto v^2$$

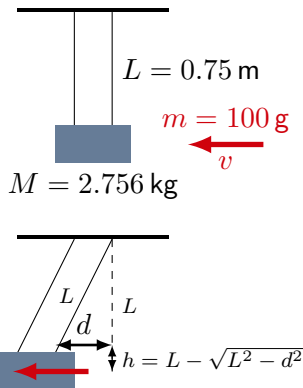
The height of the bounce goes as speed squared: the tennis ball will bounce to $9\times$ than from where it was released.



A basketball colliding with a tennis ball.

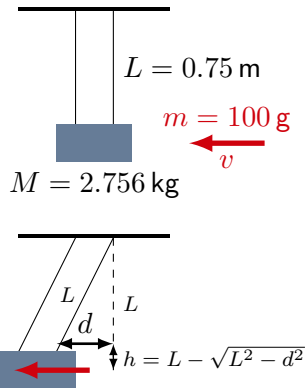
Ballistic pendulum

- The bullet/arrow embeds itself in the pendulum block.
- The pendulum rises.
- The height allows you to determine the speed of the arrow.



An arrow embeds itself into a ballistic pendulum.

Exercise: Find the speed of the arrow, given d .

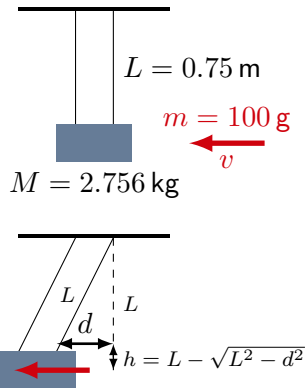


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- Conservation of momentum (inelastic collision):

$$mv = (M + m)v'$$



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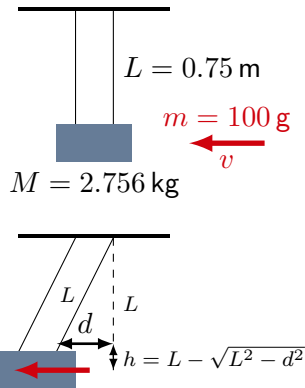
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- Conservation of energy after the collision:

$$\frac{1}{2}(M + m)v'^2 = (M + m)gh$$



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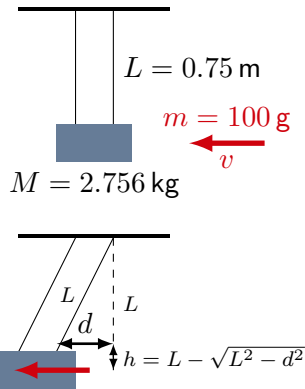
$$mv = (M + m)v'$$

- Conservation of energy after the collision:

$$\frac{1}{2}(M + m)v'^2 = (M + m)gh$$

- You should find:

$$\begin{aligned} v &= \frac{M + m}{m} \sqrt{2g \left(L - \sqrt{L^2 - d^2} \right)} \\ &= (28.56) \sqrt{(19.6 \text{ m/s}^2) \left((0.75 \text{ m}) - \sqrt{(0.75 \text{ m})^2 - d^2} \right)} \end{aligned}$$



An arrow embeds itself into a ballistic pendulum.

Centre of Mass

Momentum and Centre of Mass

Describing a system

- We can define a system of particles; that system can be particles that are freely moving with respect to each other, or it could be a solid object.
- We found that the motion of the system as a whole is determined by the external forces on the system (the internal forces cancel out).
- Want to define some quantities that allow us to describe the motion of the system as a whole:
 - Total momentum $\rightarrow$ add momenta of particles in the system.
 - Total energy (kinetic + potential) $\rightarrow$ add energies of particles in the system.
 - Position $\rightarrow$ figure out some average position of the system (the CM).

Describing a system

- The total momentum of a system:

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots$$

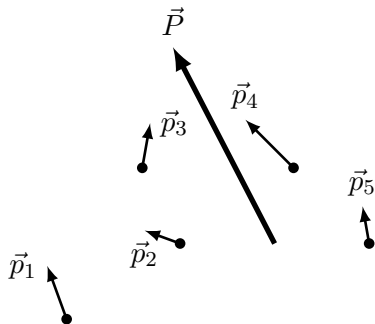
is conserved if there are no net external forces:

$$\frac{d\vec{P}}{dt} = \sum_{system} \vec{F}^{ext}$$

- Want to write Newton's 2nd Law, using the **total mass of the system, M** :

$$\sum_{system} \vec{F}^{ext} = M\vec{a}_{CM}$$

- But, a_{CM} , is the acceleration of what??



The momentum of a system of particles is given by the sum of the individual momenta of the particles.

For one particle:

$$\frac{d\vec{p}_i}{dt} = \sum \vec{F} = m\vec{a} = m\frac{d\vec{v}}{dt} = m\frac{d^2\vec{r}}{dt^2}$$

Centre of mass

- We define the centre of mass of a system to be that **location in the system that is described by:**

$$\sum_{system} \vec{F}^{ext} = M\vec{a}_{CM}$$

- That is, within the system, there exist a point, whose position, $\vec{r}_{CM}$ is such that:

$$\vec{a}_{CM} = \frac{d^2\vec{r}}{dt^2}$$

- The centre of mass is that position in the system that moves according to Newton's Second Law, where:
 - M is the total mass of the system.
 - $\sum_{system} \vec{F}^{ext}$ is the sum over external forces exerted on the system.

Location of the centre of mass

- Weighted sum of the positions of each particle in the system:

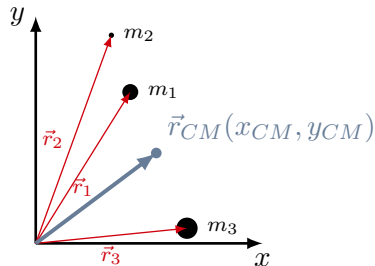
$$\vec{r}_{CM} = \frac{1}{M} \sum m_i \vec{r}_i$$

- Vector equation, so true for each component:

$$x_{CM} = \frac{1}{M} \sum m_i x_i$$

$$y_{CM} = \frac{1}{M} \sum m_i y_i$$

$$z_{CM} = \frac{1}{M} \sum m_i z_i$$



The centre of mass of particles in a system is that point of the system that moves according to Newton's Laws applied to the system as a whole.

For example:

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

Velocity and acceleration of CM

- By using the CM, we can model a system/object *as if it were a point particle* of mass M located at the centre of mass.
- The position of that particle moves with (velocity, acceleration):

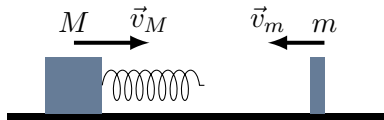
$$\vec{v}_{CM} = \frac{d}{dt} \vec{r}_{CM} = \frac{1}{M} \sum m_i \vec{v}_i$$
$$\vec{a}_{CM} = \frac{d}{dt} \vec{v}_{CM} = \frac{1}{M} \sum m_i \vec{a}_i$$

Collisions as viewed in CM frame of reference

- In the centre of mass frame of reference, the total momentum of the system is zero, since the velocity of the system in that frame is zero.

- This is sometimes useful for modeling elastic collisions:

As viewed in the CM frame, two particles are momentarily at rest at the instant when they collide and the kinetic energy is correspondingly zero at that time.



Problem from end of chapter 10. Finding the maximal compression of the spring is difficult in the lab frame. In the CM frame, it is when both blocks come to rest and all of the initial kinetic energy (evaluated in the CM frame) is equal to the potential energy stored in the spring.

Centre of mass for a continuous object

- Model a rod as being made of many mass elements, Δm , with length, Δx , each at position, x_i :

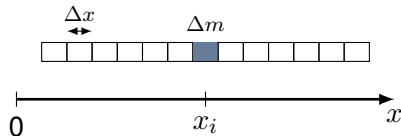
$$x_{CM} = \frac{1}{M} \sum x_i \Delta m$$

- The mass of one rod element is:

$$\Delta m = \lambda \Delta x$$

where λ is the linear mass density of the rod:

$$\lambda = \frac{M}{L}$$
$$\therefore x_{CM} = \frac{1}{M} \sum \lambda x_i \Delta x$$



To find the CM of a one dimensional object, model it as being made of many point masses, Δm , each with an infinitesimal length, Δx .

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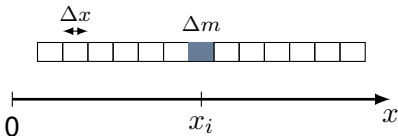
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If λ is constant:

$$x_{CM} = \frac{1}{M} \int_0^L \lambda x dx$$
$$= \frac{1}{M} \frac{1}{2} \lambda L^2 = \frac{1}{M} \frac{1}{2} \frac{M}{L} L^2 = \frac{1}{2} L$$

Centre of mass for a continuous object

$$x_{CM} = \frac{1}{M} \int_0^M x dm$$

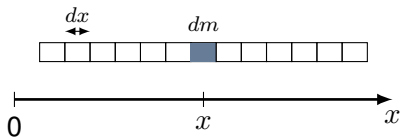
$$y_{CM} = \frac{1}{M} \int_0^M y dm$$

$$z_{CM} = \frac{1}{M} \int_0^M z dm$$

where dm is a small mass element at position (x, y, z) .

- Strategy:

- Express dm in terms of a density and a **position** coordinate.
- **Limits of the integral** correspond to range of coordinates of the object.



To find the CM of a one dimensional object, model it as being made of many point masses, Δm , each with an infinitesimal length, Δx .

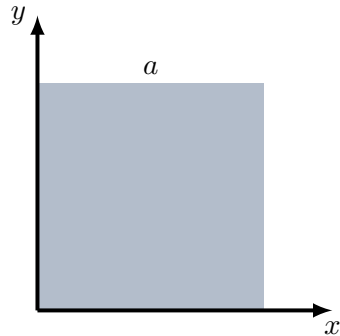
Given linear mass density, λ :

$$dm = \lambda dx \quad (\text{change of variables})$$

$$x_{CM} = \frac{1}{M} \int_0^M x dm = \frac{1}{M} \int_0^L x \lambda dx$$

$$M = \int_0^M dm = \int_0^L \lambda dx$$

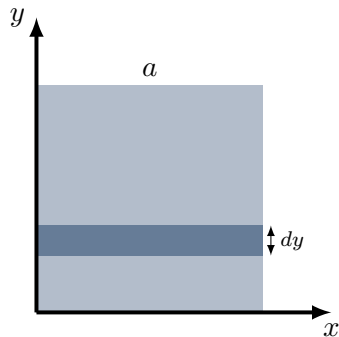
CM of continuous 2D object (square)



To find the CM of a two dimensional object, model it as being made of many 1 dimensional objects (e.g. rods) each with mass dm .

CM of continuous 2D object (square)

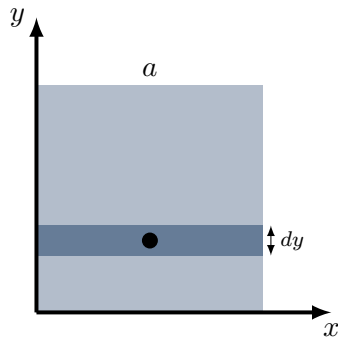
- Break up the object of dimension D into infinitesimal mass elements of dimension $D-1$:
→ A square is a sum of rods.



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CM of continuous 2D object (square)

- Break up the object of dimension D into infinitesimal mass elements of dimension $D-1$:
 - A square is a sum of rods.
- Identify the CM of the object of dimension $D-1$:
 - CM of rods is their centre.

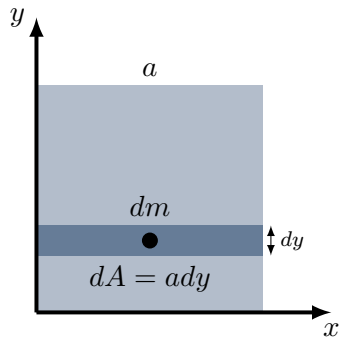


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- Break up the object of dimension D into infinitesimal mass elements of dimension $D-1$:
 - A square is a sum of rods.
- Identify the CM of the object of dimension $D-1$:
 - CM of rods is their centre.
- Write the mass of the mass element using **surface mass density**, σ :

$$dm = \sigma dA = \sigma a dy = \frac{M}{a^2} a dy = \frac{M}{a} dy$$



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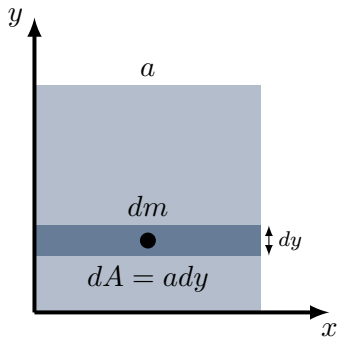
CM of continuous 2D object (square)

- Break up the object of dimension D into infinitesimal mass elements of dimension $D-1$:
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- Write the mass of the mass element using **surface mass density**, σ :

$$dm = \sigma dA = \sigma a dy = \frac{M}{a^2} a dy = \frac{M}{a} dy$$

- Solve for the CM:

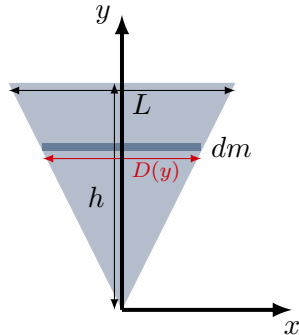
$$y_{CM} = \frac{1}{M} \int_0^M y dm = \frac{1}{M} \int_0^a y \frac{M}{a} dy = \frac{1}{2} a$$



To find the CM of a two dimensional object, model it as being made of many 1 dimensional objects (e.g. rods) each with mass dm .

CM of continuous 2D object (isosceles triangle)

- For an isosceles (L, h) triangle of mass M :
 - What is the surface mass density?
 - What is the mass of a little rod at position y ?
 - What is the integral for the CM?



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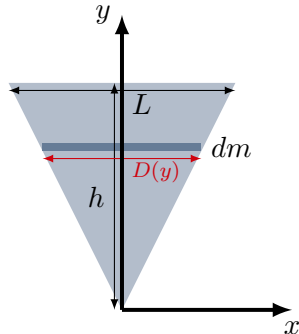
CM of continuous 2D object (isosceles triangle)

- For an isosceles (L, h) triangle of mass M :

- What is the surface mass density?

$$\sigma = \frac{M}{A} = \frac{M}{\frac{1}{2}Lh} = \frac{2M}{Lh}$$

- What is the mass of a little rod at position y ?
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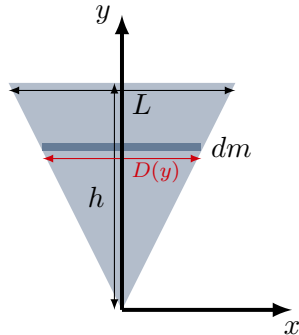
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- What is the mass of a little rod at position y ?

$$dm = \sigma dA = \sigma D(y) dy$$

$$D(0) = 0 \quad D(h) = L \quad \rightarrow \quad D(y) = \frac{L}{h}y$$

- What is the integral for the CM?



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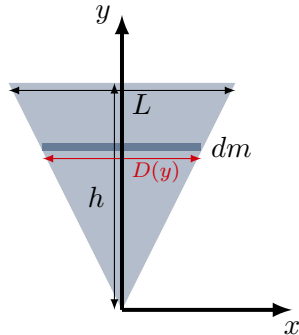
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- What is the integral for the CM?

$$y_{CM} = \frac{1}{M} \int_0^M y dm = \frac{1}{M} \int_0^h \sigma \frac{L}{h} y^2 dy$$



To find the CM of a two dimensional object, model it as being made of many 1 dimensional objects (e.g. rods) each with mass dm .

CM of continuous 3D object (parabolic volume of height, h)

- Given the volume density, ρ (don't know the mass).
- Need radius as a function of height, $r(z)$:

$$z \propto r^2 \rightarrow r(z) = a\sqrt{z}$$

- Model as small disks of radius, $r(z)$, and height, dz :

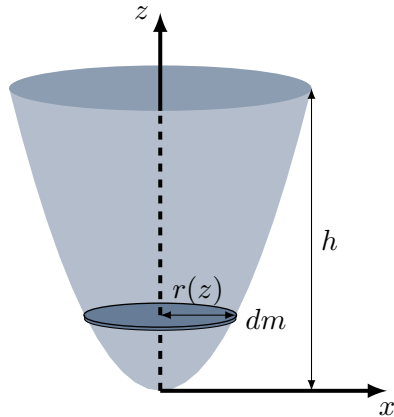
$$dm = \rho dV = \rho \pi r^2(z) dz = \rho \pi a^2 z dz$$

- The integral for the CM:

$$z_{CM} = \frac{1}{M} \int_0^M z dm = \frac{1}{M} \int_0^h \rho \pi a^2 z^2 dz$$

- The total mass:

$$M = \frac{1}{M} \int_0^M dm = \int_0^h \rho \pi a^2 z dz$$



*Model the parabolic volume as made of disks.
 y is into the page.*